Spatio-Temporal Shifting to Reduce Carbon, Water, and Land-Use Footprints of Cloud Workloads

IEEE Publication Technology, *Staff, IEEE,*

***Abstract*—In this paper, we investigate the potential of spatial and temporal shifting of cloud workloads reduce their environmental footprint. We conduct a simulation study using real-world data from multiple cloud providers (AWS and Azure) and workload traces (Spark and FaaS). Our results show that spatial shifting can lead to significant reductions in carbon, water, and land use footprints, with improvements ranging from 20% to 85%, depending on the scenario and optimization criteria. We also find that temporal shifting achieves a reduction in footprint, but to a smaller scale. Combining both methods gives the highest reductions, primarily through spatial shifting; with temporal shifting contributing a small incremental benefit. We conducted sensitivity analyses to show that our approach is resilient to prediction errors in grid mix data and across different seasons.**

***Index Terms*—Cloud computing, Sustainable development, TODO: more keywords here**

I. Introduction

The Information and Communication Technology (ICT) sector is estimated to be responsible for 1.8%–2.8% of global GHG emissions [1] and it is expected to rise in the future [2]. In particular, data centers have become one of the most energy-intensive infrastructures, with their electricity consumption projected to reach around 945 TWh globally by 2030[1]. This growth is primarily driven by data-intensive applications and services, including artificial intelligence and cryptocurrency applications.

The sustainability of cloud computing has been studied, with a primary focus on reducing carbon emissions. A wide body of work investigates carbon-aware scheduling and orchestration, exploiting spatial [3]–[7], temporal [8]–[10], and spatio-temporal [11]–[15] shifting of workloads to align demand with low-carbon electricity availability.

Recent studies have highlighted the significant environmental impacts of cloud computing beyond carbon emissions. In particular the impacts of water consumption have gained attention [16], [17]. Additionally, large-scale land use, such as one associated with data center construction,can lead to habitat disruption and biodiversity loss [18]. These factors underscore the need for a multi-dimensional approach to evaluating and optimizing the environmental footprint of cloud computing. To the best of our knowledge, land use has not been systematically integrated into sustainability-aware scheduling of cloud workloads. In this paper, we address this gap by introducing *Land Usage Effectiveness (LUE)* as a metric and incorporating it into the multi-dimensional optimization of workload placement.

In this paper, building on the idea proposed in our previous work [19], we investigate spatial and temporal shifting of cloud workloads as strategies to reduce carbon, water, and land use footprints. We conduct an extensive simulation study using real-world data from major public cloud providers (Azure and AWS) and representative workloads (FaaS and big data analytics), demonstrating the potential of multi-dimensional sustainability-aware scheduling.

In summary, our main contributions are:

- We propose a holistic formulation for workload scheduling that optimizes carbon, water, and land use impacts, leveraging temporal and spatial cloud workload shifting
- We provide a systematic evaluation based on real-world workload traces and grid mix data, quantifying trade-offs under spatial, temporal, and spatio-temporal shifting scenarios. Moreover, we conduct sensitivity studies on workload flexibility, grid mix prediction errors, and seasonal variations.
- We release all code and data from this study to support transparency, reproducibility, and further research.[2]

The rest of the paper is organized as follows. Section II provides background on data center environmental impacts and workload flexibility. Section III presents our system model and problem formulation. It also describes our methodology, including the proposed optimization algorithms. Section V presents our simulation setup, including a detailed description of the scenarios characteristics such as workload types, data center specifications, and grid datasets. Section VI presents the results of our empirical evaluation and discusses interesting insights and implications of our findings. Section VIII reviews related work in carbon-aware computing and sustainability in cloud computing. Finally, Section IX concludes the paper and outlines directions for future research.

II. Background - Data Centers impacts

A. Operation

Data centers are among the most energy-demanding facilities globally. They require large amounts of electricity to power servers, cooling systems, and backup power supplies. This energy usage results in significant GreenHouse Gas (GHG) emissions, especially in regions where electricity is generated from fossil fuels. In fact, the environmental impact of data centers is directly tied to the energy mix of their respective grids. The energy mix directly influences the **Carbon Intensity (CI)**, which measures the greenhouse gas emissions per unit of electricity consumed (typically expressed in gCO_2e/kWh); the **Energy Water Intensity Factor (EWIF)**,

[1]IEA - Energy demand from AI [Online]

[2]Repository available at: TODO: add link

which captures the amount of water withdrawn or consumed during electricity generation (measured in liters/kWh), which varies with the cooling technologies; and the **Energy Land Intensity Factor (ELIF)**, which quantifies the land use required for producing electricity (expressed in m_2/kWh). An example of these intensities' estimates are reported in [20]–[22]. These are also the main sources used in our experiments (see Table VI), and in the discussion section we will analyze how these coefficients relates to each other.

However, the grid energy mix is not the only factor affecting the environmental impact of data centers. The efficiency of data center operations also depends on the infrastructure and technologies used.

The **Power Usage Effectiveness (PUE)**[3], is defined as the ratio of the total energy consumed by a datacenter facility to the energy consumed by its IT equipment. A PUE of 1.0 indicates perfect efficiency, where all energy is used exclusively for computation.

Moreover, the efficiency of cooling systems, which are essential to prevent overheating of servers, can significantly affect water usage. The choice of cooling technology (e.g., air cooling vs. liquid cooling) can lead to different water consumption. The **Water Usage Effectiveness (WUE)**[4], measures the water consumption of a datacenter relative to its IT energy usage, typically expressed in liters of water per kilowatt-hour (l/kWh).

Finally, since these infrastructures occupy a significant amount of land, the **Land Usage Effectiveness (LUE)**[5] should be accounted as well. To the best of our knowledge, this metric has not been standardized and it has not been used previously. Similarly to PUE and WUE, the LUE metric aims to quantify the physical land footprint of data centers in relation to IT energy consumption, capturing the spatial resource requirements of large-scale infrastructure. Thus, for the scope of our work, we define it as the ratio of the total land occupied by the data center property and the IT energy consumption, and thus it is expressed in square meters per kilowatt-hour (m^2/kWh).

B. Construction and Dismantling

Data center facilities also have a significant environmental impact for what concerns the construction and eventual dismantling. Large amounts of raw materials like concrete, steel, copper, and rare earth elements are required for their construction. Moreover, these facilities also occupy significant land. The level of the ecological footprint of a data center building can be formally certified, e.g. with the LEED certification scheme[6]. Furthermore, the dismantling of data centers generates substantial e-waste, which often contains toxic materials, like lead, nickel, and mercury, which, if not properly recycled, can leach into soil and water, ultimately having an adverse impact on human health and the environment [23]. The carbon associated with the extraction, processing, and transportation of construction materials, as well as with the production of IT equipment (known as embodied carbon) accounts for a share of the total carbon footprint of a data center. However, in this work, we consider only the share of operational carbon emissions of the workloads.

TABLE I
NOTATION

Symbol	*Description*
t_0, H, δ_t, $\mathcal{T}$	period start, end, step duration and set of steps
d, $\mathcal{D}$	region/datacenter and set of regions/datacenters
PUE_d	region PUE
WUE_d	region WUE
LUE_d	region LUE
A_d	region land occupation area
IT_d	region IT energy usage
$\mathcal{C}_d(t)$	region carbon footprint
$\mathcal{W}_d(t)$	region water footprint
$\mathcal{L}_d(t)$	region land footprint
$CI^e_{G_d}(t)$	region Energy Carbon Intensity prediction
$EWIF^e_{G_d}(t)$	region Energy Water Intensity Factor prediction
$ELIF^e_{G_d}(t)$	region Energy Land Intensity Factor prediction
$p_d(t)$	data center footprint profile at time t
$p^{norm}_d(t)$	normalized data center footprint profile at time t
$p^{norm}_{global}(t)$	global footprint profile
v, $\mathcal{V}$	VM instance and set of VM instances
CPU_v, MEM_v	VM number of CPUs and RAM size
$P_{v,j}$	VM v power draw for job j
$P^{CPU}_{v,j}$	VM's CPU power draw based on job CPU utilization
$Watts_{min}$	minimum CPU power draw (fixed for one scenario)
$Watts_{max}$	maximum CPU power draw (fixed for one scenario)
P^{MEM}_v	VM's RAM power draw
B_v	VM's network bandwidth
j, $\mathcal{J}$	job and set of jobs
o_j, a_j, t_j	job's arrival location and arrival and deadline time
s_j , $\mathcal{D}[j]$	job's data size and region availability
v_j , n_j	job's configuration (VM instance and nr. of nodes)
u_j, r_j	job's expected mean utilization and expected runtime
$C(j, d, t)$	scheduling footprints
$M(j, d, t)$	migration/data transfer footprints
NEI	Network Energy Intensity
$L(j, d)$	migration/data transfer latency
$E(j, d, t)$	execution footprint

III. PROBLEM STATEMENT

In this section we formalize the problem.

A. System Model

First of all, let us consider the temporal horizon H, and let us discretize the time. Thus, let us define $\mathcal{T} = [t_0, t_0 + \delta_t, t_0 + 2\delta_t, ..., H]$, as the set of time steps. Let us define also $\mathcal{D}$ as the set of regions/datacenters, and consider a data center d and a time t. To evaluate its impact accounting for multiple factors (i.e. carbon, water and land use), we define its sustainability profile as follow:

$$\forall t \in \mathcal{T},\ d \in \mathcal{D} \quad \mathbf{p}_d(t) = \begin{pmatrix} \mathcal{C}_d(t) \\ \mathcal{W}_d(t) \\ \mathcal{L}_d(t) \end{pmatrix} \tag{1}$$

where:

[3] The Green Grid: PUE: A Comprehensive Examination of the Metric [Online]
[4] The Green Grid: WP#35 - Water Usage Effectiveness (WUE): A Green Grid Data Center Sustainability Metric [Online]
[5] K. Sumrani - Decarbonization: From Data Centers to Sustainable Infrastructure [Online]
[6] What Are the Sustainable Data Center Standards and Certification? [Online]

- $\mathcal{C}_d(t)$ is the predicted carbon intensity of datacenter d from time t to time $t + \delta_t$ and it is defined as
$$CI^e_{G_d}(t) \times PUE_d \quad (2)$$
It is measured in $\frac{gCO_2}{kWh}$ and represents the carbon emitted per kWh consumed. We consider the carbon intensity prediction of the regional grid of data center d at time t, namely $CI^e_{G_d}(t)$, where e is the predictive model. This intensity is multiplied by the Power Usage Effectiveness of the data center, WUE_d, to account for the overall energy usage of a datacenter for each kWh of computation.
- $\mathcal{W}_d(t)$ is the predicted water intensity and it is defined as
$$WUE_d + EWIF^e_{G_d}(t) \times PUE_d \quad (3)$$
It is measured in $\frac{l}{kWh}$ and represents the water consumed per kWh considering the average Energy Water Intensity Factor prediction of the regional grid, $EWIF^e_{G_d}(t)$ and the Water Usage Effectiveness of the data center, WUE_d.
- $\mathcal{L}_d(t)$ is the predicted land use intensity and it is defined as
$$LUE_d + ELIF^e_{G_d}(t) \times PUE_d \quad (4)$$
It is measured in $\frac{m^2}{kWh}$, it is the amount of land used per kWh considering the Energy Land Intensity Factor prediction of the regional grid at time t, $ELIF^e_{G_d}(t)$ and the Land Usage Effectiveness, defined as $LUE_d = \frac{A_d}{P^{IT}_d}$, where A_d is the total area of land occupied by the datacenter property and P^{IT}_d is the energy used to power the IT equipment.

The aforementioned energy intensity factors of the electric grid intensity depends on the energy mix, which is both time and space dependent. Thus, let us define S as the list of sources in the energy mix, namely $\{Solar, Wind, Hydro, Geothermal, Biomass, Nuclear, Coal, Gas, Oil, Unknown\}$, and let $MIX^e_{G_d,s}(t)$ be the share of source $s \in S$ in the mix prediction of G_d. Thus, we can define the energy intensity factors as:

- $CI^e_{G_d}(t) = \sum_{s \in S} MIX^e_{G_d,s}(t) \times CI_s$
- $EWIF^e_{G_d}(t) = \sum_{s \in S} MIX^e_{G_d,s}(t) \times EWIF_s$
- $ELIF^e_{G_d}(t) = \sum_{s \in S} MIX^e_{G_d,s}(t) \times ELIF_s$

where CI_s, $EWIF_s$ and $ELIF_s$ are specified in table VI.

Moreover, let us define $\mathcal{V}$ as the set of virtual machine instances, for each $v \in \mathcal{V}$ it is known the power draw P_v defined as:

$$P_{v,j} = CPU_v \times P^{CPU}_{v,j} + RAM_v \times P^{MEM} \quad (5)$$

where: CPU_v is the number of CPUs, RAM_v: is the size of RAM expressed in GB, B_v is the network bandwidth expressed in Gbps, P^{MEM} is the memory power drawn and $P^{CPU}_{v,j} = Watts_{min} + u_j \times (Watts_{max} - Watts_{min})$ is the CPU power draw, both defined according to the cloud carbon footprint methodology[7].

Finally, let us define also the set of job requests J. For each request $j \in J$ it is known: the arrival/origin region/location $o_j \in D$, the regions in which data is available $\mathcal{D}[j] \subseteq \mathcal{D}$, data size $s_j \in \mathbb{N}$, number of nodes $n_j \in \mathbb{N}$, the expected utilization $u_j \in \mathbb{N}$, the VM instance $v_j \in \mathcal{V}$, the expected runtime $r_j \in \mathbb{Z}$, the arrival time $a_j \in \mathcal{T}$, and the deadline $t_j \in \mathcal{T}$.

We aim to optimize footprints associated with the job requests scheduling, considering both data migration and execution, by leveraging spatio-temporal shifting approaches.

[7]Cloud Carbon Footprint: Methodology [Online]

IV. Methodology

In this section, we present our methodology. We define an evaluation method for the footprints associated with a job request. Then we present the scheduling methods considered in our evaluation. Finally we present our experiments design.

A. Evaluating environmental footprints

Firstly, we need a methodology to evaluate the footprints of executing a request in a certain region at a certain time. Let us consider we want to predict the associated footprint of computing a job request j in region/datacenter d at time t. If the data related to the request is not available in d, data migration is required, and it is necessary to account for its environmental impact.

To account for migration footprint overhead, data transmission energy is estimated as follows:

$$M(j,d,t) = \begin{cases} s_j \times NEI & \text{if } d \notin \mathcal{D}[j] \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where NEI kWh/GB is the Network Energy Intensity, as proposed in [24].

To account for the execution footprint we need to know at what time its computation will start. We assume that if data is not migrated the execution start would be at arrival time, since the scheduling execution time is negligible. If the data needs to be migrated we need to compute the data migration latency as follow:

$$L(j,d) = \begin{cases} \frac{s_j}{B_{v_j}} & \text{if } d \notin \mathcal{D}[j] \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

To account for the execution footprint, we define

$$E(j,d,t_i) = \int_{t_i+L(j,d)}^{t_i+L(j,d)+r_j} P_j \cdot \mathbf{p}^{norm}_d(t)\, dt, \quad (8)$$

where $\mathbf{p}^{norm}_d(t) \in \mathbb{R}^3$ is the normalized profile vector of region d for carbon, water, and land use. We define the normalization operator applied componentwise.

$$\mathcal{N}(\mathbf{p}_d(t)) = \frac{\mathbf{x} - \min_{d,t} \mathbf{p}_d(t)}{\max_{d,t} \mathbf{p}_d(t) - \min_{d,t} \mathbf{p}_d(t)}$$

Thus the regional normalized profile is $\mathbf{p}^{norm}_d(t) = \mathcal{N}(\mathbf{p}_d(t))$ We shall consider global intensities to account for migration footprint, and we define them as the normalized (over time) average intensities, namely: $\mathbf{p}^{norm}_{global}(t) = \mathcal{N}(\operatorname{avg}_d \mathbf{p}_d(t))$. Finally, we can compute the overall environmental cost of scheduling the job.

$$C(j,d,t_i) = (E(j,d,t_i) + M(j,n,t_i) \times \mathbf{p}^{norm}_{global}(t)) \times \theta \quad (9)$$

where θ is the weight vector to linearly combine the three impact factors, namely: $\theta = \begin{pmatrix} w_{\mathcal{C}} & w_{\mathcal{W}} & w_{\mathcal{L}} \end{pmatrix}$.

B. Scheduling Methods

We consider the following shifting methods:

- **Spatial Shifting**: jobs should be executed where their overall impact is minimized (considering data migration when required).

$$d^*, t^* = argmin_{d \in \mathcal{D}} C(j, d, a_j),\ a_j \tag{10}$$

- **Temporal Shifting**: jobs should be executed when their overall impact is minimized.

$$d^*, t^* = o_j,\ argmin_{t \in [a_t, d_t - r_t, \delta_t]} C(j, o_j, t) \tag{11}$$

- **Spatio-Temporal Shifting**: optimize both when and where the jobs' overall impact is minimized.

$$d^*, t^* = argmin_{d \in \mathcal{D},\ t \in [a_t, d_t - (r_t + L(j,d)), \delta_t]} C(j, d, t) \tag{12}$$

For methods including spatial shifting, we assume that once data is transferred to another region, the origin location persistently store the data. This is because for periodic jobs we leverage the fact that data is available in multiple regions without transferring it multiple times. As a baseline we consider a method in which jobs are executed where the request originates, namely the **Local** baseline:

$$d^*, t^* = o_j,\ a_j \tag{13}$$

C. Experiments design

In this work we focus on two types of workloads. We have selected two workloads: a user-facing, latency-sensitive workload (FaaS) and a delay-tolerant, batch-processing workload (big data). For each workload type, we considered real traces and derived two real-world scenarios to simulate their scheduling and execution.

a) Simulation overview: We implemented a discrete time simulation. Algorithm 1 details the simulation procedure. At each time step we update the sustainability profiles of data centers and new job requests ready to be scheduled. Then each job can be scheduled according to the shifting approach previously defined. Finally, since we assume that once data is transmitted to a region it is persistently stored there, the data availability set of the job is updated for future scheduling.

Algorithm 1 Simulation Procedure

Input: $\mathcal{D}$ data centers, `scheduling_method()` $\in \{(10), (11), (12), (13)\}$
for $t \in \mathcal{T}$ **do**
  Get sustainability profiles $p_d(t)$ for each $d \in \mathcal{D}$
  Get jobs ready to be scheduled at time t, $J(t)$.
  for j in $J(t)$ **parallel do**
    $d^*, t^* =$ `scheduling_method()`
    if `scheduling_method()` $\in \{(10), (12)\}$ **then**
      $\mathcal{D}[j] = \mathcal{D}[j] \cup \{d^*\}$
    end if
  end for
end for
Output: Final schedule over horizon H

b) Scenario 1 - Azure and FaaS: The first scenario is based on Azure regions and a FaaS workload. The regions selected for this scenario are "swedencentral", "southcentralus", "centralus" and "eastus". Table II reports regions characteristics. PUEs and WUEs values where taken from Azure issued sustainability reports[8]. The land occupation data is taken from online available sources (swedencentral [9], southcentralus [10], centralus [11], eastus2 [12]).

As for the workload, we chose the 2019 Azure Functions dataset[13]. The source data is composed of two sets of daily CSV files: one detailing per-minute invocation counts for each function, and another providing daily summary statistics of function execution times, including averages and key percentiles (0th, 1st, 25th, 50th, 75th, 99th, and 100th). We utilised the first seven days of this dataset to synthesise 100,000 requests of function execution a day. The selection of a function for each request was probabilistic and weighted by the function's total daily invocation count, such that more frequently invoked functions appeared more often in the trace, mirroring real-world usage patterns.

Upon selecting a function, we sampled a synthetic runtime by performing linear interpolation on its known daily execution time percentiles. The arrival time for the request, resolved to the minute-of-day, was similarly sampled from the function's empirical per-minute invocation distribution. Finally, each request was assigned a datacenter location from a predefined set. A single virtual machine configuration (2 vCPU and 4 GB of RAM) is considered for this scenario, as specified in the paper [25].

For this scenario we compare only the **Local** baseline (L) and the **Spatial Shifting** (SP) approach. We assume that that the functions are tolerant to the latency required to change regions but not to be delayed in time. Thus, for this reason, we exclude **Temporal Shifting** (T) and **Spatio-temporal Shifting** (STP) scheduling methods. Thus, deadlines are defined for this scenario.

TABLE II
AZURE DATA BY REGION SCENARIO 1.

Region	Location	PUE	WUE	Land Occup. (sqm)
swedencentral	Sweden	1.172	0.16	1300000
southcentralus	Texas	1.307	1.82	43664
centralus	Iowa	1.16	0.19	37904
eastus2	Virginia	1.144	0.17	102193

c) Scenario 2 - AWS and Big Data: The second scenario is based on AWS and Big Data workloads. The regions selected for this scenario are: "us-east-1", "us-west-1", "eu-central-1", "eu-west-2", and "eu-north-1". Table IV reports

[8]Microsoft, Sustainability Fact Sheets - Sweden [Online], Texas [Online], Iowa [Online], Virginia [Online]

[9]Invest Stockholm, "Two municipalities in the Stockholm region sell land to Microsoft," [Online].

[10]Datacenters.com, "Microsoft Azure: South Central US-Texas," [Online].

[11]Datacenters.com, "Microsoft Azure: Central US-Iowa," [Online].

[12]Datacenters.com, "Microsoft Azure: East US–Virginia (Boydton, VA) Data Center," [Online].

[13]Github: Azure Functions Trace 2019 [Online]

TABLE III
AWS VM INSTANCE TYPES AND SPECIFICATIONS

Type	vCPU	Mem (GB)	Bandwidth (Mbps)
c. l, xl, xxl	2, 4, 8	4, 8, 16	500, 750, 1000
m4. l, xl, xxl	2, 4, 8	8, 18, 34	450, 750, 1000
r4. l, xl, xxl	2, 4, 8	16, 32, 65	450, 850, 1700

regions characteristics. PUEs and WUEs values were taken from AWS website [14].

The land occupation data is estimated based on the 10-k form released be AWS in 2024 [15], in which it is stated that 454546 sqmt where owned/rented by AWS for datacenters. The estimation is also based on declared availability zones for each region. Namely, estimated land occupation for a given region is calculated by taking the number of availability zones (AZs) in that region, dividing it by the total number of AZs (117), and then multiplying by the total land occupation of all data centers.

As for the workload, we synthetically generate requests from available experimental Spark traces[16]. We randomly generated 50k requests over a week. For the arrival pattern, we generated 50% of requests as periodic (in line with literature analysis: about 43% [26], about 60% [27]) and the rest as ad-hoc jobs. We generate synthetic requests for each day of the selected week. The arrival of ad-hoc jobs follows a Poisson distribution with $\lambda = target_{np}/n_{days} * minutesperday \approx 2.5$. The periodic jobs first instance arrival time is randomly picked within the first 12 hours and their periodicity is set uniformly at random among $\{2, 4, 8, 12\}$ hours. For each request, the origin location is picked uniformly at random from the regions considered for this scenario. The number of nodes and the VM instance and the expected runtime are the ones reported in the traces. Table III summarizes the VM instances considered in our experiments and their specifications are set accordingly (memory, cpu [17], and bandwidth [18]).

For this scenario, we consider all of the scheduling methods defined above. We also introduce a variation of **Spatial Shifting**, in which we do not persistently store data transferred when migrating (S). For **Temporal Shifting** (T) and **Spatio-Temporal Shifting** (TSP), the deadline is either its periodicity or, for ad hoc jobs, a tunable delay tolerance parameter.

TABLE IV
AWS DATA BY REGION SCENARIO 2.

Region	Location	PUE	WUE	Land Occup. (sqm)
us-east-1	Virginia	1.15	0.12	233101
us-west-1	California	1.17	0.51	116550
eu-central-1	Frankfurt	1.35	0.01	116550
eu-west-2	London	1.11	0.04	116550
eu-north-1	Stockholm	1.10	0.02	116550

[14] AWS Cloud: Increasing Efficiency [Online]
[15] Amazon.com Inc. - Amazon 2024 Annual Report [Online]
[16] Github: scout: A Tool for Cloud Configuration Optimization [Online]
[17] Amazon EC2 Instance types [Online]
[18] Specifications for Amazon EC2 previous generation instances [Online]

V. EVALUATION

In this section we provide the experimental setup, data sources and simulation parameters.

A. Experimental Setup

a) Hardware and Software Environment: The experiments were conducted on the following system:

- **CPU:** Intel(R) Core(TM) i9-10920X CPU @ 3.50GHz
- **RAM:** 260 GB
- **Operating System:** Ubuntu 24.04.3 LTS
- **Simulation Framework:** Custom, implemented in python.

b) Grid data and intensities: We collected data from various sources. Since we considered multiple regions, we collected publicly available data from different grid operators (sw [19], uk [20], de[21], ercot [22], miso [23], pjm [24], caiso [25]). The historical mix data is used to compute the actual environmental impact of the workload execution, while to make decisions noised mix data is used to simulate the prediction of grids' mix. The data sources are not standardized, thus we had to harmonize the data to have a common format. We considered a granularity of 1 hour and finer-grained data were reshaped considering the mean value for each hour. Table V depicts the relationship between grids and cloud providers' regions.

TABLE V
DATA SOURCES FOR GRID MIXES

EU	sw	uk	de	
	eu-north-1 swedencentral	eu-west-2	eu-central-1	
USA	ercot	miso	pjm	caiso
	southcentralus	centralus	us-east-1 eastus2	us-west-1

To incorporate uncertainties into the grid mix data, we introduce synthetic noise. Specifically, for each time step and region, the renewable share is perturbed by an error term ϵ_r sampled from a normal distribution $\mathcal{N}(0, \sigma_r)$, where $\sigma_r = mae \cdot \frac{1}{\sqrt{2/\pi}}$ and mae is the target mean absolute error. The error ϵ_r is then distributed among the renewable sources (solar, wind, geothermal, hydro, biomass) according to a probability distribution designed to reflect their prediction difficulty: solar gets 45% of error, since it is the hardest source to estimate [28]; wind gets 30% [29]; both hydro [30] and geothermal [31] get 10%; and biomass get 5% of the error, being the most controllable among the renewable sources. Each renewable source i receives a noise $\epsilon_i = \epsilon_r \cdot w_i$, where w_i is the weight for source i. This approach is a tentative to simulates realistic prediction errors and reflects the varying uncertainty associated with each energy source.

[19] ENTSOE Transparency Platform [Online]
[20] National Electricity System Operator API [Online]
[21] SMARD - Download market data [Online]
[22] EIA - Wholesale Electricity Market Data - ERCOT [Online]
[23] EIA - Wholesale Electricity Market Data - MISO[Online]
[24] EIA - Wholesale Electricity Market Data - PJM [Online]
[25] EIA - Wholesale Electricity Market Data - CAISO [Online]

For non-renewable sources, a uniform noise is applied to each share to maintain the total mix balance.

Finally, in our simulations, to compute the average intensities of the mixes, we consider the intensity coefficients reported in Table VI. Carbon intensity values correspond median of the lifecycle emissions reported in IPCC Annex III, Table IPCC A.III.2 table (page 1335) [20]. Except for the oil carbon intensity, which is missing in such source, then, we consider the coefficient provided by [32].Water intensity values were obtained form the Water Consumption Factors presented in Tables 1 and 2 (pages 12-13) of the NREL technical report on energy-related water consumption [21]. Land use intensity values were extracted from a comprehensive a survey [22]. The survey does not provide a value for oil, but the authors assume it to be comparable to gas, so we did as well. For all the factors, when multiple options were presented for the same energy source, we considered the average of median intensity values. Note that we could not find any source for the oil EWIF, thus we assume the mean of the other sources' intensity values. As well as for the the unknown share.

TABLE VI
IMPACT COEFFICIENTS FOR VARIOUS ELECTRICITY GENERATION TECHNOLOGIES. **SOURCES:** CARBON: IPCC [20], OUR WORLD IN DATA [32] (OIL). WATER: NREL [21]. LAND USE [22].

Energy Source	**CI** $gCO2e/kWh$	**EWIF** l/kWh	**ELIF** ha/TWh
Nuclear	12	1.957	7.1
Geothermal	38	5.827	45
Biomass	485	1.145	29065
Coal	820	1.8	1000
Wind	11.5	0	6065
Solar	38.67	1.385	1650
Hydro	24	17	650
Gas	490	1.099	1155
Oil	720	n/a (3.776)	1155
Unknown	n/a (293.24)	n/a (3.776)	n/a (4532)

c) Other parameters: For both scenarios we considered one-week horizon and varied the week in different seasons, namely: winter (t_0 = 2024-01-15T00:00:00Z, H = 2024-01-22T00:00:00Z), spring (t_0 =2024-04-15T00:00:00Z, H = 2024-04-22T00:00:00Z),summer (t_0 =2024-07-15T00:00:00Z, H = 2024-07-22T00:00:00Z), autumn (t_0 =2024-10-15T00:00:00Z, H = 2024-10-22T00:00:00Z). And the step δ_t is set to one minute.

To account for the uncertainty in the prediction of the workload, we considered different values of MAE, namely: 0.05, 0.1, 0.15, 0.2. For statistical significance, each simulation is repeated 6 times with different random seeds, namely: 0, 1, 2, 3, 4, 5.

The factors weights are set to: [1.0, 0.0, 0.0] only carbon footprint is optimized, [0.0, 1.0, 0.0] only water footprint is optimized, [0.0, 0.0, 1.0] only land footprint is optimized, [0.333, 0.333, 0.334] all footprints are optimized equally.

Moreover, to account for the energy consumption of the network, we considered a network energy intensity of 0.06 kWh/GB [33]. The memory power draw P^{MEM}, and the minimum and maximum CPU power draw $(Watts_{min}, Watts_{max})$ are respectively set to 0.357 Watt/GB and (0.74, 3.5) for AWS and (0.78, 3.76) for Azure [26]. Finally, we set P_d^{IT} to the estimate provided by EIA. Assuming the IT power draw to be 100 MW we assumed an IT power consumption of $8760 * 100 * 10^3$ kWh over one year.

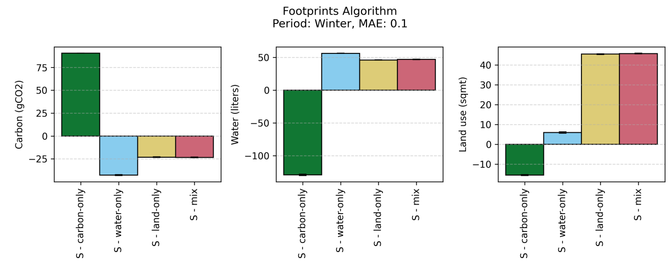

Fig. 1. Spatial shifting - Azure - percent improvement over local baseline - Winter week, mae = 0.1

VI. RESULTS

In this section, we present the results of our experiments.

A. Scenario 1: Azure + FaaS

For this first scenario, we present the results of the spatial shifting approach optimizing for all factor weight combinations.

a) Spatial Shifting: Figure 1 shows the footprint improvement with respect to the local baseline of spatial shifting. This experiment considered the winter week and an error of 10% on the renewable share prediction. We compare all factor weights' combinations defined above.

The results demonstrate that regional shifting of FaaS workloads across Azure regions leads to noticeable improvements in carbon efficiency. The plots show that when optimising for a particular metric (carbon, water, or land use footprint), we can achieve a $45 - 85\%$ improvement in that metric. When considering the metric for which the optimisation was performed, we see that the carbon footprint benefits the most, with up to an 85% reduction in emissions compared to the local baseline. This significant reduction reflects the negligible migration cost of FaaS workloads. A reduction of about 85% in carbon footprint leads to an increase of about 40% in water footprint. The land use footprint also increases, but to a lesser extent, having observed about a 25% increase.

Similarly, optimizing for water footprint (about 50% reduction) leads to a significant increase in carbon footprint (about 125% increase), but the land use footprint decreases proportionally to the water footprint reduction (about 50% reduction). On the other hand optimizing for land use footprint (about 45% reduction) leads to a moderate increase in carbon footprint (about 15% increase) and a small reduction in water footprint (less than 10% reduction).

Figure 3 shows the request distribution across the regions. To help us understand this distribution, let us consider Figure 2, which simultaneously reports intensity values over the winter week and the efficiency metrics for each factor for each region.

[26]Cloud Carbon Footprint: Methodology [Online]

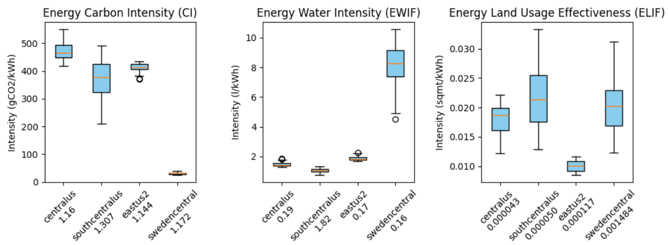

Fig. 2. Grid intensities - Azure - Winter week, mae = 0.1

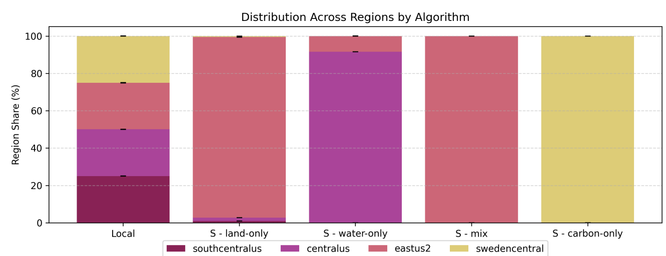

Fig. 3. Spatial shifting - Azure - Request distribution across different regions for different criteria - Winter week, mae = 0.1

When optimising for carbon, all the requests are scheduled to "centralus" which is not the most energy efficient region (PUE 1.172) and it has the grid with the lowest carbon intensity. When optimising for water, a small percentage of requests going to "eastus2" and most of requests are scheduled to "centralus", which is not the most water efficient region (WUE 0.19) nor is it the one with the smallest grid water intensity. However, it is the region with the lowest impact on water when considering both data center efficiency and grid intensity. Finally, when optimising for land use, nearly all the requests are scheduled to "eastus2", which even if it is not the region with the least land occupation, its grid has a lower land use intensity.

b) Sensitivity analysis: To understand how seasonality affects the effectiveness of the scheduling methods, we perform a sensitivity analysis considering weeks in the four seasons: winter, spring, summer, and autumn. For this analysis, we use a fixed error for the grid's renewable share prediction (mae = 0.1). On the other hand, to understand how the prediction error affects the effectiveness of the scheduling methods, we perform a sensitivity analysis considering different values of prediction error (mae = $\{0.05, 0.1, 0.15, 0.2\}$). We use a fixed week in winter for this analysis.

From Table VII it is possible to notice that the prediction error leads to very small differences, almost flat. Indeed, the results indicate that the improvement in reduction in all three metrics remains robust even for higher error levels.

On the other hand, it seems that the effectiveness of spatial shifting varies with seasonal changes in grid intensities, but the overall trend of improvement remains consistent. Land use factors, in particular, demonstrate varying impacts across different seasons (37%-57%).

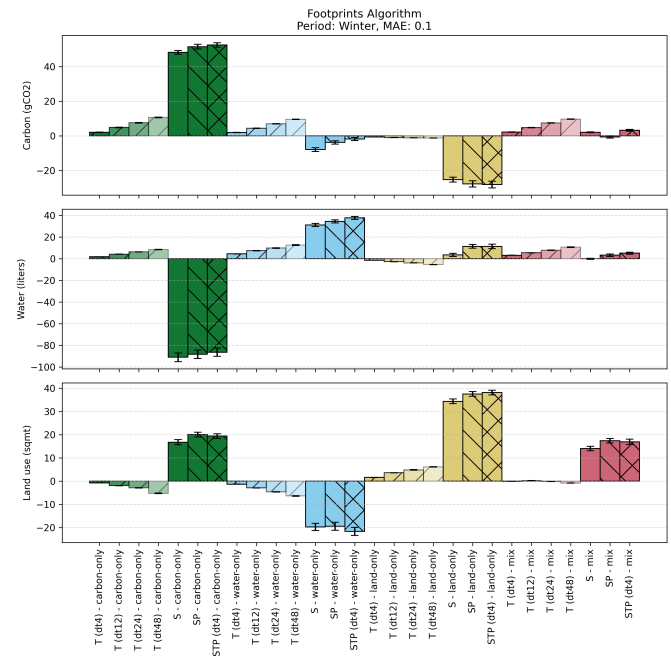

Fig. 4. Aws - Percent improvement over local baseline - Winter week, mae = 0.1

B. Scenario 2: Aws + Big Data

For this second scenario, not only we investigate the spatial shifting method, but also temporal shifting and spatio-temporal shifting. We consider the same week in winter as in Scenario 1. We consider a prediction error (mae) of 0.1. We consider the same factor weights' combinations as in Scenario 1. For temporal shifting we consider different values of delay tolerance (dt) = $\{4, 12, 24, 48\}$ hours.

a) Baseline Footprints: The baseline algorithm (Local) has about 17.500 kilograms of carbon footprint, 200.000 liters of water footprint and just above 2000 square meters of land use footprint. In these graphs we show the improvement in percentage of the other algorithms with respect to the baseline. Figure 4 shows the results of our experiments.

b) Temporal Shifting: Firstly, we can see that the temporal shifting approach achieves relatively modest improvements compared to spatial shifting, with reductions ranging from 6% to 12% even at the highest delay tolerance of 48 hours.

c) Spatial Shifting: The spatial shifting approach (SP) achieves a significant improvement in each metric when optimising for it. In this case the improvements range from $20 - 45\%$ footprint reductions over the baseline. Figure 5 show the request distribution across the regions. To help us understand this distribution, let us consider Figure 6, which simultaneously reports intensity values over the winter week and the efficiency metrics for each factor for each region.

When optimising for carbon, about 50% of requests are scheduled to "eu-north-1" which is the most energy efficient region (PUE 1.1) and also the region with the grid with lowest carbon intensity. When optimising for water, about 40% of requests are scheduled to "us-west-2" which is not the most water efficient region (WUE 0.04) but it has the grid with

TABLE VII
AZURE - SENSITIVITY ANALYSIS (LEFT: SEASON, RIGHT: MIX RENEWABLE SHARE PREDICTION ERROR)

algorithm	period	carbon improvement		water improvement		land use improvement		mae	carbon improvement		water improvement		land use improvement	
		mean	std	mean	std	mean	std		mean	std	mean	std	mean	std
S - carbon-only	autumn	91.28	0.03	-118.26	0.75	-8.10	0.46	0.05	90.55	0.03	-129.41	1.12	-15.46	0.26
	spring	86.56	0.04	-139.90	1.17	18.29	0.32	0.1	90.55	0.03	-129.41	1.12	-15.46	0.26
	summer	91.87	0.02	-123.28	1.15	-13.85	0.29	0.15	90.55	0.03	-129.41	1.12	-15.46	0.26
	winter	90.55	0.03	-129.41	1.12	-15.46	0.26	0.2	90.55	0.03	-129.41	1.12	-15.46	0.26
S - water-only	autumn	-48.74	0.55	55.33	0.22	6.57	0.35	0.05	-43.64	0.52	56.60	0.23	4.95	0.32
	spring	-29.97	0.39	59.40	0.19	-11.63	0.41	0.1	-42.75	0.52	56.42	0.23	5.99	0.32
	summer	-24.66	0.39	50.82	0.21	32.46	0.13	0.15	-42.81	0.51	56.25	0.23	6.79	0.32
	winter	-42.75	0.52	56.42	0.23	5.99	0.32	0.2	-41.36	0.51	55.84	0.23	8.59	0.30
S - land-only	autumn	-25.17	0.50	40.90	0.30	57.00	0.17	0.05	-23.40	0.43	47.12	0.28	45.71	0.18
	spring	-15.63	0.35	34.32	0.37	42.24	0.22	0.1	-23.12	0.42	46.16	0.27	45.40	0.19
	summer	-13.88	0.41	39.31	0.36	37.54	0.12	0.15	-22.33	0.42	44.14	0.29	44.85	0.18
	winter	-23.12	0.42	46.16	0.27	45.40	0.19	0.2	-22.03	0.42	42.84	0.30	44.15	0.19
S - mix	autumn	-25.86	0.49	42.35	0.27	57.16	0.17	0.05	-23.34	0.43	47.10	0.28	45.74	0.18
	spring	-23.35	0.35	49.22	0.24	42.73	0.22	0.1	-23.34	0.43	47.10	0.28	45.74	0.18
	summer	-17.64	0.40	49.55	0.23	38.33	0.12	0.15	-23.34	0.43	47.10	0.28	45.74	0.18
	winter	-23.34	0.43	47.10	0.28	45.74	0.18	0.2	-23.32	0.43	47.08	0.28	45.57	0.18

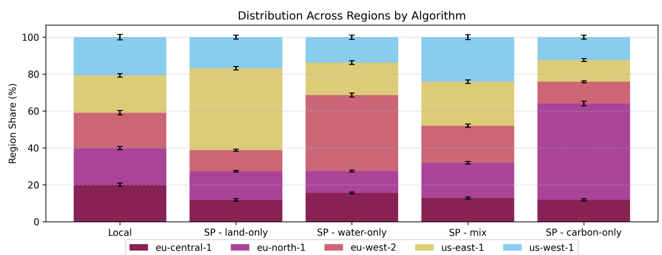

Fig. 5. Spatial shifting - AWS - Request distribution across different regions for different criteria - Winter week, mae = 0.1

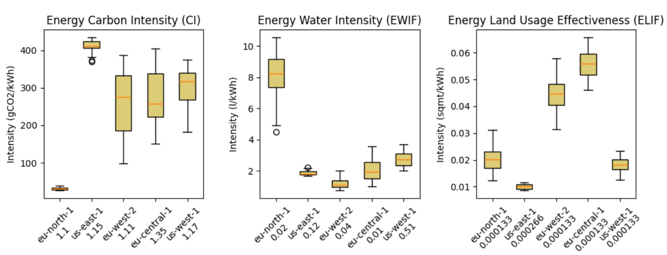

Fig. 6. Grid intensities - Aws - Winter week, mae = 0.1

lowest impact on water. Finally, when optimising for land use, about 45% of requests are scheduled to "us-east-1", which even if is the region with the most land occupation, it is the one which grid has the lowest impact on land use.

d) Migration overheads of Spatial Shifting: For this scenario we consider also an alternative approach to spatial shifting (S). In this approach the data associated with each request must be transferred each time a migration occurs, even if the data was transferred to that region previously. Results shows that this approach still acheives significant improvements over the baseline, but SP acheives about $5-6\%$ better results in the optimised metric.

Table VIII provides a summary of the migration overhead for the three factors, it shows the percentage of footprints caused by migration. The migration overhead varies based on the scheduling algorithm and the weights used, and it is relatively low, ranging from 0.85% to 12.67% for carbon, 0.89% to 5.47% for water, and 1.01% to 7.68% for land use. The overhead of SP is about half compared to S.

TABLE VIII
SPATIAL SHIFTING (PERSISTENT STORAGE VS NON-PERSISTENT STORAGE) - AWS - AVERAGE PERCENTAGE OF MIGRATION FOOTPRINTS - WINTER WEEK, MAE = 0.1

	Carbon (%)		*Water (%)*		*Land Use (%)*	
	S	SP	S	SP	S	SP
carbon-only	12.67	6.96	3.45	1.80	7.68	4.13
water-only	3.50	1.85	5.47	2.94	3.07	1.58
land-use-only	3.51	1.80	4.55	2.60	6.62	3.65
mix	1.69	0.85	1.67	0.89	1.89	1.01

e) Spatio-temporal Shifting: Finally, the best outcome is achieved by the spatio-temporal shifting approach (STP) with delay tolerance 4 hours, which results in 55% improvement in carbon footprint when optimizing for carbon, about 40% improvement in water footprint when optimizing for water and about 48% improvement in land use footprint when optimizing for land use. However, these improvements with respect to only spatial shifting are small.

f) Sensitivity analysis: Following the same approach used for scenario 1 we conduct sensitivity analysis for this scenario as well. Table IX shows the footprint improvements over the local baseline for different seasons and for different prediction errors for the grid renewable share (mae). The results fluctuates slightly with the prediction error (2-3%) but the improvements remain significant even with a high prediction error of 0.2. Also the season does not affect significantly the results, with variations of 4% at most considering the optimized factor.

VII. DISCUSSION

A. Scenario 1: Azure + FaaS

The results demonstrate that regional shifting of FaaS workloads across Azure regions leads to noticeable improvements in carbon efficiency. This significant reductions reflects the

TABLE IX
AWS - SENSITIVITY ANALYSIS (LEFT: SEASON, RIGHT: MIX RENEWABLE SHARE PREDICTION ERROR)

algorithm	period	carbon improvement		water improvement		land use improvement		mae	carbon improvement		water improvement		land use improvement	
		mean	std	mean	std	mean	std		mean	std	mean	std	mean	std
SP carbon-only	autumn	52.00	1.33	-65.81	2.96	21.26	0.86	0.05	51.72	1.32	-88.67	3.86	20.31	0.97
	spring	45.96	1.11	-68.89	2.95	30.19	0.89	0.1	51.60	1.32	-88.18	3.86	20.05	1.00
	summer	51.05	1.34	-70.26	3.13	29.75	0.90	0.15	51.27	1.34	-86.49	3.91	19.29	0.99
	winter	51.60	1.32	-88.18	3.86	20.05	1.00	0.2	49.97	1.25	-81.37	3.67	16.97	0.95
SP water-only	autumn	-3.02	0.88	32.83	1.16	-18.67	1.57	0.05	-3.61	0.95	34.71	1.46	-20.22	1.65
	spring	-0.53	0.81	32.77	1.32	-20.88	1.64	0.1	-3.81	0.96	34.42	1.46	-19.40	1.72
	summer	-27.04	2.13	29.59	1.20	9.16	1.38	0.15	-4.27	1.00	34.16	1.44	-19.20	1.71
	winter	-3.81	0.96	34.42	1.46	-19.40	1.72	0.2	-4.04	1.04	34.17	1.44	-18.63	1.71
SP land-only	autumn	-25.84	2.05	11.83	1.63	41.04	1.02	0.05	-28.87	1.87	12.51	1.87	37.73	1.01
	spring	-21.40	1.81	-3.26	1.57	38.43	0.98	0.1	-27.72	1.81	11.42	1.76	37.49	1.00
	summer	-23.65	1.46	2.72	1.45	39.74	1.00	0.15	-27.32	1.71	10.60	1.63	37.32	1.01
	winter	-27.72	1.81	11.42	1.76	37.49	1.00	0.2	-25.64	1.57	8.49	1.71	36.78	0.95
SP mix	autumn	1.99	0.90	7.43	1.13	24.13	0.78	0.05	0.17	0.56	1.58	1.03	17.69	1.20
	spring	2.78	1.24	15.98	1.67	16.26	1.20	0.1	-0.59	0.64	3.33	1.13	17.39	0.99
	summer	6.49	0.92	-0.22	1.33	20.48	1.08	0.15	0.48	0.81	2.75	1.09	16.43	1.40
	winter	-0.59	0.64	3.33	1.13	17.39	0.99	0.2	-6.59	1.11	8.37	1.43	19.93	1.29
T (dt4) carbon-only	autumn	5.09	0.26	3.51	0.10	0.59	0.09	0.05	2.67	0.15	1.55	0.07	-0.89	0.04
	spring	4.28	0.28	2.63	0.16	1.37	0.11	0.1	2.02	0.14	1.90	0.08	-0.79	0.03
	summer	4.60	0.23	2.00	0.12	1.46	0.17	0.15	1.92	0.13	1.39	0.09	-0.62	0.03
	winter	2.02	0.14	1.90	0.08	-0.79	0.03	0.2	1.72	0.11	1.04	0.06	-0.47	0.04
T (dt4) water-only	autumn	4.93	0.24	6.81	0.09	0.32	0.11	0.05	2.15	0.12	4.85	0.11	-1.60	0.06
	spring	4.62	0.27	6.24	0.16	0.51	0.10	0.1	1.91	0.11	4.49	0.11	-1.33	0.05
	summer	4.35	0.24	7.26	0.16	0.99	0.13	0.15	1.76	0.11	4.34	0.11	-1.29	0.06
	winter	1.91	0.11	4.49	0.11	-1.33	0.05	0.2	1.84	0.11	4.11	0.11	-1.19	0.05
T (dt4) land-only	autumn	1.97	0.23	-0.15	0.13	3.71	0.12	0.05	-0.48	0.04	-1.05	0.06	2.10	0.05
	spring	2.67	0.24	0.48	0.17	3.75	0.14	0.1	-0.65	0.04	-1.41	0.05	1.62	0.05
	summer	2.11	0.20	-0.21	0.09	4.80	0.20	0.15	-0.53	0.05	-0.68	0.05	1.40	0.05
	winter	-0.65	0.04	-1.41	0.05	1.62	0.05	0.2	-0.08	0.07	-0.51	0.05	1.22	0.05
T (dt4) mix	autumn	5.54	0.25	5.49	0.09	1.71	0.13	0.05	2.50	0.13	3.49	0.11	-0.07	0.05
	spring	5.67	0.29	4.87	0.18	2.74	0.12	0.1	2.22	0.13	3.19	0.11	-0.10	0.03
	summer	5.24	0.26	5.77	0.15	3.12	0.17	0.15	2.18	0.13	2.47	0.11	-0.14	0.04
	winter	2.22	0.13	3.19	0.11	-0.10	0.03	0.2	1.95	0.12	2.43	0.10	0.00	0.04
STP (dt4) carbon-only	autumn	54.46	1.30	-63.54	3.00	21.24	0.87	0.05	52.86	1.31	-86.21	3.86	19.38	1.02
	spring	48.67	1.13	-72.15	3.29	31.87	0.91	0.1	52.59	1.31	-86.23	3.87	19.37	1.02
	summer	53.37	1.34	-71.52	3.22	30.97	0.93	0.15	52.52	1.32	-86.48	3.87	19.46	1.03
	winter	52.59	1.31	-86.23	3.87	19.37	1.02	0.2	52.24	1.32	-86.26	3.87	19.30	1.02
STP (dt4) water-only	autumn	1.66	0.83	37.05	1.13	-19.98	1.42	0.05	-1.09	0.87	38.27	1.41	-23.27	1.74
	spring	3.93	0.73	36.43	1.31	-23.77	1.63	0.1	-1.79	0.93	37.66	1.38	-21.60	1.72
	summer	-22.21	1.75	33.30	1.18	6.84	1.07	0.15	-2.58	0.89	37.30	1.42	-21.25	1.69
	winter	-1.79	0.93	37.66	1.38	-21.60	1.72	0.2	-2.06	0.93	37.11	1.45	-20.60	1.79
STP (dt4) land-only	autumn	-23.46	2.02	10.23	1.51	42.53	0.99	0.05	-28.82	1.84	12.47	1.87	38.60	0.97
	spring	-22.95	2.00	0.64	1.34	40.20	0.96	0.1	-28.16	1.93	11.35	2.01	38.11	1.01
	summer	-22.05	1.57	2.83	1.15	41.86	1.00	0.15	-28.63	1.93	12.43	1.94	38.17	1.01
	winter	-28.16	1.93	11.35	2.01	38.11	1.01	0.2	-27.82	1.85	11.59	1.84	38.08	1.02
STP (dt4) mix	autumn	9.21	1.09	10.30	1.35	24.10	0.82	0.05	2.88	0.46	4.36	0.76	18.20	1.13
	spring	12.29	0.55	18.50	1.66	17.32	1.09	0.1	3.27	0.49	5.09	0.88	16.91	1.17
	summer	12.86	0.78	3.76	1.42	20.38	0.87	0.15	5.33	0.59	3.57	0.85	15.25	1.09
	winter	3.27	0.49	5.09	0.88	16.91	1.17	0.2	0.60	0.56	7.95	1.31	16.76	1.20

negligible migration cost of FaaS workloads. Overall, these results highlight the potential of spatial shifting significantly enhancing the sustainability of FaaS workloads by leveraging regional variations in grid intensities. Moreover, based on the distribution of requests across regions, it is evident that the energy mix of the grid plays a more crucial role in determining the environmental impact than the data center efficiency metrics. However, variations in the data center efficiency metrics still are decisive in determining the optimal region for workload allocation when the energy mix differences are not substantial.

B. Scenario 2: Aws + Big Data

The results highlight the potential of spatial shifting significantly enhancing the sustainability by leveraging, for the most part, regional variations in grid intensities. The migration overhead remains minimal. This indicates that the benefits of spatial shifting in reducing the overall footprint significantly outweigh the footprints associated with migration. Combined spatial and temporal shifting can lead to even more significant improvements, although the additional benefits are relatively modest compared to spatial shifting alone. On the other hand, even if temporal shifting can lead to some improvements, these are relatively modest compared to spatial shifting. This suggests that while temporal shifting can contribute to sustainability goals, its impact is limited by the inherent variability of renewable energy availability within a single region. Though, it is worth noting that even if temporal shifting has less savings, it does not require to move data between regions, which sometimes will not be possible due to security/privacy

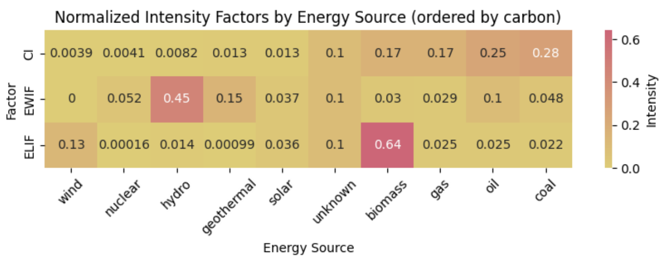

Fig. 7. Energy Intensity Factors normalized

concerns.

C. Comparison of the two scenarios

From the results we can see that with diverse providers and workloads, the results are different but the improvements are relevant in both. In particular, we can see that for Azure we obtain the highest improvements, since there is no footprint overhead when migrating, while for AWS there is a small overhead. However, this overhead is such that the improvements are still significant. Moreover, it allows us to obtain a more balanced distribution of requests among the regions.

D. Trade-offs analysis

As expected, in both scenarios, optimising for one metric can lead to trade-offs in the other metrics. For instance, optimising for carbon footprint lead to a significant increase in water footprint, as already observed in previous studies [17].

This is particularly evident when optimizing for carbon, which results in a significant increase in water footprint. Optimizing for water results in an increase in both carbon and land use footprints, but to a lesser extent compared to the increase of water footprint when optimizing only for carbon. Optimizing for the mix of the three factors results in a balanced reduction of all three footprints, but the improvements are lower compared to optimizing for only one factor.

To help us understand these trade-offs, we consider the energy intensity factors for each energy source. Figure 7 shows the energy intensity factors normalized between 0 and 1 (by dividing by the sum of each factor across all sources). It seems that carbon and water are negatively correlated (e.g. hydro, coal), but not all low carbon intensity sources have high water intensity (e.g., wind and nuclear). On the other hand, land use seems to not be correlated with the other factors, although there are sources with high land use and low water intensity (e.g. biomass), and sources with low land use and high carbon and water intensity (e.g. oil).

E. Threats to validity

This study presents some threats to validity that should be considered when interpreting the results.

a) Resource availability: we assume that there always are resources available, which might not be true in reality. However, considering public cloud providers, this is a fair assumption.

b) Generality: the scenarios considered may not fully represent the diversity of real-world cloud environments, as we focused on specific providers (Azure and AWS) and workload types (FaaS and big data). Moreover, the workload traces used for the AWS scenario may not be representative of production cloud workloads since arrival patterns are synthetically randomized.

c) Input quality: the accuracy of our results is contingent upon the quality of the input data, including grid mix predictions that in our case are synthetically generated from the historical data using an arbitrary noise distribution. Morever, to estimate the LUE parameter, we use estimations for the region IT power consumption and the land occupation, since precise data is available. This, this estimates could affect the reliability of our findings.

d) Spatial shifting constraints: our models assume that workloads can be freely shifted across regions and time without considering potential constraints such as latency requirements, data sovereignty issues, or user experience impacts.

e) Economic costs: our simulations do not account for potential economic costs associated with shifting workloads, which could influence the feasibility of implementing such strategies in practice.

VIII. Related Work

Enhancing resource management and operational efficiency, cloud orchestration has fueled a wide body of research [34]. Allocating resources, both optimizing energy consumption and carbon awareness [3]–[5], [8], [10]–[12], [14], [15] have received a significant interest. Recently, the impact on water resources of computation has raised some concerns, but few studies have focused on this issue/aspect [16], [17]. Differently from previous works, we included also land use into an approach that assesses and optimizes the overall impact.

A. Carbon-Aware Computing

In cloud computing, carbon-aware optimisation focuses on aligning workload execution with variations in the carbon intensity of the electricity supply over time and/or across different geographical areas.

a) Temporal Shifting: Hanafy et al. [8] propose CarbonScaler, where cloud batch jobs dynamically adjust server allocations based on grid carbon intensity, and demonstrates the performance of this approach in a Kubernetes prototype.

Wiesner et al. [9] analyse the temporal shifting benefits considering different regions, demonstrating the carbon savings potential of delay-tolerant workloads in Germany, Great Britain, France, and California.

Piontek et al. [10] propose a Kubernetes scheduler that shifts non-critical jobs in time so to reduce carbon emissions based on their prediction algorithm.

b) Spatial Shifting: Gao et al. [3] provide a scheduler that controls the traffic directed to each data center optimizing a 3way-trade-off between access latency, electricity cost, and carbon footprint.

Maji et al. [4] propose load balancing in VMware's Avi Global Server Load Balancer, which uses a linear scoring function to select the optimal data center in terms of marginal carbon intensity and the distance between the client and the data center.

Chadha et al. [5] present GreenCourier, a Kubernetes scheduler designed to reduce carbon emissions associated with serverless functions scheduled across geographically distributed regions.

Lindberg et al. [6] propose a metric, named locational marginal carbon emission, to guide spatial shifting.

Zheng et al. [7] evaluate the idea of migrating workloads between data centers to reduce curtailment. It has been shown that migrating workloads from fossil-fuel-powered grids to renewable-energy-powered grids during curtailment hours simultaneously reduces emissions and absorbs excess renewable generation.

c) Spatio-Temporal Shifting: Zhou et al. [11] use Lyapunov optimization to perform load balancing across geo-distributed data centers, capacity right-sizing (turning off idle servers), and server speed scaling (adjusting the CPU frequency). The objective here is to optimize a 3way-trade-off between electricity cost, SLA (Service-Level Agreement) requirements, and emission reduction budget.

Souza et al. [12] propose a provisioner that minimizes both the number of active servers and the associated carbon emission.

Sukprasert et al. [13] analyses the benefits and limitations of carbon-aware spatiotemporal shifting for cloud workloads.

Cordingly et al. [14] propose a prototype for computing resource aggregation that minimizes the carbon footprint of a serverless application through carbon-aware load distribution.

Lin et al. [15] proposes a coordinated scheduling framework that aligns data center capacity planning with grid dynamics in both spatial and temporal dimensions to avoid increasing carbon emissions, energy costs, and reducing grid reliability.

B. Beyond Carbon-Aware Computing

Recent research has started to address the environmental impact of cloud and AI workloads beyond just carbon emissions. Li et al. [16] highlight the hidden water footprint of AI models, uncovering the significant amounts of freshwater consumed for both cooling and electricity generation. Their study estimates that training large AI models results in hundreds of thousands of liters of direct water consumption.

Jiang et al. [17] introduce WaterWise, a job scheduling framework that co-optimizes carbon and water sustainability for geographically distributed data centers by employing a Mixed Integer Linear Programming (MILP)-based scheduler, which dynamically shifts workloads to regions with optimal carbon and water conditions while considering delay tolerance constraints. This work aligns with ours, which also emphasizes the need for multi-dimensional sustainability metrics when optimizing cloud workload allocation, highlighting the need for comprehensive sustainability-aware orchestration strategies. Of the literature, [17] is the closest to our work. However, we focus more on regional differences beyond grid intensities. For example, we consider real-world regional-specific PUE values instead of considering them constant across regions and set to an arbitrary value (1.2). Furthermore, rather than adopting the wet-bulb temperature-based method used by WaterWise to estimate WUE values, we use the WUE values declared by providers. As shown in our previous work [19], this method can lead to estimations that differ substantially from the values declared by the providers.

Although carbon and water footprints have been studied, land use has not been considered in cloud sustainability research. Therefore, the question of how to incorporate land use considerations remains an open research challenge.

C. Comparison with own previous work

We first presented the idea of holistic sustainable scheduling at the 1st International Workshop on Low Carbon Computing [19]. In this extended journal, we address some of the limitations of our previous work, such as non-realistic assumptions. For this reason, we no longer consider live migration, but instead we focus on providing an experimental evaluation of spatial shifting. Moreover, to ensure the validity of our evaluation, we use real-world workload traces instead of synthetic data to simulate jobs. Furthermore, we include temporal shifting which was not considered in our previous work.

IX. Conclusion

To conclude, we summarize the key contributions of our study highlighting the obtained results and the limitations and threats to validity of our work.

In this paper we have shown that spatial and temporal shifting can significantly reduce the environmental footprint of cloud workloads. Spatial shifting, in particular, has demonstrated substantial improvements across all three environmental factors considered (carbon, water, and land use), with reductions ranging from 20% to 85% depending on the scenario and optimization criteria. Temporal shifting also contributes to footprint reduction, to a lesser extent, with improvements up to 12%. The combination of both methods yields the highest reductions, although the incremental benefit over spatial shifting alone is small.

Our sensitivity analyses indicate that these methods are robust to prediction errors in grid mix data and maintain their effectiveness across different seasons. Finally, we have demonstrated that it is possible to reduce the environmental footprint and mitigate trade-off implications by optimising all the factors simultaneously using a mixed strategy. Overall, our findings highlight the potential of a more sustainable workload scheduling as a practical approach considering diverse cloud providers and workloads.

Furthermore, our simulations could incorporate other aspects such as cross-region dependability, load balancing requirements, failure tolerance, and low-latency delivery for worldwide customers.

Acknowledgments

This work was supported by the Engineering and Physical Sciences Research Council under grant number UKRI154 ("Casper: Carbon-Aware Scalable Processing in Elastic Clusters" TODO: Ask Novella

References

[1] C. Freitag, M. Berners-Lee, K. Widdicks, B. Knowles, G. S. Blair, and A. Friday, "The real climate and transformative impact of ict: A critique of estimates, trends, and regulations," *Patterns*, vol. 2, no. 9, p. 100340, 2021. [Online]. Available: https://doi.org/10.1016/j.patter.2021.100340

[2] L. Belkhir and A. Elmeligi, "Assessing ict global emissions footprint: Trends to 2040 & recommendations," *Journal of cleaner production*, vol. 177, pp. 448–463, 2018. [Online]. Available: https://doi.org/10.1016/j.jclepro.2017.12.239

[3] P. X. Gao, A. R. Curtis, B. Wong, and S. Keshav, "It's not easy being green," in *Proceedings of the ACM SIGCOMM 2012 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communication*, ser. SIGCOMM '12. New York, NY, USA: Association for Computing Machinery, 2012, p. 211–222. [Online]. Available: https://doi.org/10.1145/2342356.2342398

[4] D. Maji, B. Pfaff, V. P R, R. Sreenivasan, V. Firoiu, S. Iyer, C. Josephson, Z. Pan, and R. K. Sitaraman, "Bringing carbon awareness to multi-cloud application delivery," in *Proceedings of the 2nd Workshop on Sustainable Computer Systems*, ser. HotCarbon '23. New York, NY, USA: Association for Computing Machinery, 2023. [Online]. Available: https://doi.org/10.1145/3604930.3605711

[5] M. Chadha, T. Subramanian, E. Arima, M. Gerndt, M. Schulz, and O. Abboud, "Greencourier: Carbon-aware scheduling for serverless functions," in *Proceedings of the 9th International Workshop on Serverless Computing*, ser. WoSC '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 18–23. [Online]. Available: https://doi.org/10.1145/3631295.3631396

[6] J. Lindberg, Y. Abdennadher, J. Chen, B. C. Lesieutre, and L. Roald, "A guide to reducing carbon emissions through data center geographical load shifting," in *Proceedings of the Twelfth ACM International Conference on Future Energy Systems*, ser. e-Energy '21. New York, NY, USA: Association for Computing Machinery, 2021, p. 430–436. [Online]. Available: https://doi.org/10.1145/3447555.3466582

[7] J. Zheng, A. A. Chien, and S. Suh, "Mitigating curtailment and carbon emissions through load migration between data centers," *Joule*, vol. 4, no. 10, pp. 2208–2222, 2020. [Online]. Available: https://doi.org/10.1016/j.joule.2020.08.001

[8] W. A. Hanafy, Q. Liang, N. Bashir, D. Irwin, and P. Shenoy, "Carbonscaler: Leveraging cloud workload elasticity for optimizing carbon-efficiency," *Proc. ACM Meas. Anal. Comput. Syst.*, vol. 7, no. 3, Dec. 2023. [Online]. Available: https://doi.org/10.1145/3626788

[9] P. Wiesner, I. Behnke, D. Scheinert, K. Gontarska, and L. Thamsen, "Let's wait awhile: how temporal workload shifting can reduce carbon emissions in the cloud," in *Proceedings of the 22nd International Middleware Conference*, ser. Middleware '21. New York, NY, USA: Association for Computing Machinery, 2021, p. 260–272. [Online]. Available: https://doi.org/10.1145/3464298.3493399

[10] T. Piontek, K. Haghshenas, and M. Aiello, "Carbon emission-aware job scheduling for kubernetes deployments," *Journal of supercomputing*, vol. 80, pp. 549–569, 2023. [Online]. Available: https://doi.org/10.1007/s11227-023-05506-7

[11] Z. Zhou, F. Liu, R. Zou, J. Liu, H. Xu, and H. Jin, "Carbon-aware online control of geo-distributed cloud services," *IEEE Transactions on Parallel and Distributed Systems*, vol. 27, no. 9, pp. 2506–2519, 2016. [Online]. Available: https://doi.org/10.1109/TPDS.2015.2504978

[12] A. Souza, S. Jasoria, B. Chakrabarty, A. Bridgwater, A. Lundberg, F. Skogh, A. Ali-Eldin, D. Irwin, and P. Shenoy, "Casper: Carbon-aware scheduling and provisioning for distributed web services," in *Proceedings of the 14th International Green and Sustainable Computing Conference*, ser. IGSC '23. New York, NY, USA: Association for Computing Machinery, 2024, p. 67–73. [Online]. Available: https://doi.org/10.1145/3634769.3634812

[13] T. Sukprasert, A. Souza, N. Bashir, D. Irwin, and P. Shenoy, "On the limitations of carbon-aware temporal and spatial workload shifting in the cloud," in *Proceedings of the Nineteenth European Conference on Computer Systems*, ser. EuroSys '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 924–941. [Online]. Available: https://doi.org/10.1145/3627703.3650079

[14] R. Cordingly, J. Kaur, D. Dwivedi, and W. Lloyd, "Towards serverless sky computing: An investigation on global workload distribution to mitigate carbon intensity, network latency, and cost," in *2023 IEEE International Conference on Cloud Engineering (IC2E)*, 2023, pp. 59–69. [Online]. Available: https://doi.org/10.1109/IC2E59103.2023.00015

[15] L. Lin and A. A. Chien, "Adapting datacenter capacity for greener datacenters and grid," in *Proceedings of the 14th ACM International Conference on Future Energy Systems*, ser. e-Energy '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 200–213. [Online]. Available: https://doi.org/10.1145/3575813.3595197

[16] P. Li, J. Yang, M. A. Islam, and S. Ren, "Making ai less 'thirsty'," *Commun. ACM*, vol. 68, no. 7, p. 54–61, Jun. 2025. [Online]. Available: https://doi.org/10.1145/3724499

[17] Y. Jiang, R. B. Roy, R. Kanakagiri, and D. Tiwari, "Waterwise: Co-optimizing carbon- and water-footprint toward environmentally sustainable cloud computing," in *Proceedings of the 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming*, ser. PPoPP '25. New York, NY, USA: Association for Computing Machinery, 2025, p. 297–311. [Online]. Available: https://doi.org/10.1145/3710848.3710891

[18] T. Newbold, L. N. Hudson, S. L. Hill, S. Contu, I. Lysenko, R. A. Senior, L. Börger, D. J. Bennett, A. Choimes, B. Collen *et al.*, "Global effects of land use on local terrestrial biodiversity," *Nature*, vol. 520, no. 7545, pp. 45–50, 2015. [Online]. Available: https://doi.org/10.1038/nature14324

[19] G. Attenni and N. Bartolini, "Environmentally-conscious cloud orchestration considering geo-distributed data centers," 2025. [Online]. Available: https://arxiv.org/abs/2507.11563

[20] IPCC, "Climate change 2014: Mitigation of climate change. annex iii: Technology-specific cost and performance parameters," 2014. [Online]. Available: https://www.ipcc.ch/site/assets/uploads/2018/02/ipcc_wg3_ar5_annex-iii.pdf

[21] J. Macknick, R. Newmark, G. Heath, and K. Hallett, "Consumptive water use for u.s. power production," 2011. [Online]. Available: https://www.nrel.gov/docs/fy11osti/50900.pdf

[22] J. Lovering, M. Swain, L. Blomqvist, and R. R. Hernandez, "Land-use intensity of electricity production and tomorrow's energy landscape," *PLoS One*, vol. 17, no. 7, p. e0270155, 2022. [Online]. Available: https://doi.org/10.1371/journal.pone.0270155

[23] V. N. Pinto, "E-waste hazard: The impending challenge," *Indian journal of occupational and environmental medicine*, vol. 12, no. 2, pp. 65–70, 2008. [Online]. Available: doi={https://doi.org/10.4103/0019-5278.43263}

[24] J. Aslan, K. Mayers, J. G. Koomey, and C. France, "Electricity intensity of internet data transmission: Untangling the estimates," *Journal of Industrial Ecology*, vol. 22, no. 4, pp. 785–798, 2018. [Online]. Available: https://doi.org/10.1111/jiec.12630

[25] M. Shahrad, R. Fonseca, I. Goiri, G. Chaudhry, P. Batum, J. Cooke, E. Laureano, C. Tresness, M. Russinovich, and R. Bianchini, "Serverless in the wild: Characterizing and optimizing the serverless workload at a large cloud provider," in *2020 USENIX annual technical conference (USENIX ATC 20)*, 2020, pp. 205–218. [Online]. Available: https://www.usenix.org/conference/atc20/presentation/shahrad

[26] J. Panneerselvam, L. Liu, and N. Antonopoulos, "Characterisation of hidden periodicity in large-scale cloud datacentre environments," in *2017 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData)*, 2017, pp. 496–503. [Online]. Available: https://doi.org/10.1109/iThings-GreenCom-CPSCom-SmartData.2017.79

[27] S. A. Jyothi, C. Curino, I. Menache, S. M. Narayanamurthy, A. Tumanov, J. Yaniv, R. Mavlyutov, I. n. Goiri, S. Krishnan, J. Kulkarni, and S. Rao, "Morpheus: towards automated slos for enterprise clusters," in *Proceedings of the 12th USENIX Conference on Operating Systems Design and Implementation*, ser. OSDI'16. USA: USENIX Association, 2016, p. 117–134. [Online]. Available: https://www.usenix.org/conference/osdi16/technical-sessions/presentation/jyothi

[28] A. Alfadda, R. Adhikari, M. Kuzlu, and S. Rahman, "Hour-ahead solar pv power forecasting using svr based approach," in *2017 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*. IEEE, 2017, pp. 1–5. [Online]. Available: https://doi.org/10.1109/ISGT.2017.8086020

[29] I. Okumus and A. Dinler, "Current status of wind energy forecasting and a hybrid method for hourly predictions," *Energy Conversion and Management*, vol. 123, pp. 362–371, 2016. [Online]. Available: https://doi.org/10.1016/j.enconman.2016.06.053

[30] F. Zayas-Gato, A. Díaz-Longueira, P. Arcano-Bea, Á. Michelena, J. L. Calvo-Rolle, and E. Jove, "A new deep learning-based approach for predicting the geothermal heat pump's thermal power of a real bioclimatic house," *Applied Intelligence*, vol. 55, no. 6, pp. 1–20, 2025. [Online]. Available: https://doi.org/10.1007/s10489-025-06457-7

[31] M. Bilgili, S. Keiyinci, and F. Ekinci, "One-day ahead forecasting of energy production from run-of-river hydroelectric power plants with a

deep learning approach," *Scientia Iranica*, vol. 29, no. 4, pp. 1838–1852, 2022. [Online]. Available: https://doi.org/10.24200/sci.2022.58636.5825

[32] Our World in Data, "What are the safest and cleanest sources of energy?" 2020. [Online]. Available: https://ourworldindata.org/safest-sources-of-energy

[33] J. Aslan, K. Mayers, J. G. Koomey, and C. France, "Electricity intensity of internet data transmission: Untangling the estimates," *Journal of Industrial Ecology*, vol. 22, no. 4, pp. 785–798, 2018. [Online]. Available: https://doi.org/10.1111/jiec.12630

[34] D. Weerasiri, M. C. Barukh, B. Benatallah, Q. Z. Sheng, and R. Ranjan, "A taxonomy and survey of cloud resource orchestration techniques," *ACM Comput. Surv.*, vol. 50, no. 2, may 2017. [Online]. Available: https://doi.org/10.1145/3054177

X. Biography Section

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated `\includegraphics` command within an optional argument. (You can create your own custom macro containing the `\includegraphics` command to make things simpler here.)

If you include a photo:

Michael Shell Use `\begin{IEEEbiography}` and then for the 1st argument use `\includegraphics` to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

If you will not include a photo:

John Doe Use `\begin{IEEEbiographynophoto}` and the author name as the argument followed by the biography text.